

Assignment 3

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```
#####
##### Data preparation
#####

# The lines below are run if the packages were not yet installed.
# Otherwise, they are commented out so the code is not executed.
# install.packages("pastecs")
# install.packages("reshape2")
# install.packages("tidyverse")

# The pastecs package contains functions like `stat.desc` and `by` to compute
# descriptive statistics and stratify data based on a specified categorical
# variable.
library(pastecs)
# The reshape2 package contains the function `melt` to convert data from a wide
# format to a long format.
library(reshape2)
# The tidyverse package contains functions for data wrangling and
# generating figures, such as `ggplot`, among others.
library(tidyverse)

# The working directory is retrieved and stored as an object. This assumes that
# the local repository on the user's machine is used.
wd_path <- getwd()

# The data file path is stored as an object based on where the data is stored
# in the repository and the file name.
data_path <- paste0(wd_path, "/TextMessages.csv")

# The data is loaded into the RStudio environment.
# Since the file has a .csv extension, the function `read.csv` is used.
# An object is created which contains the imported data from the specified
# file path `data_path`.
text_data <- read.csv(data_path, header=TRUE)

# The command `names` calls the data frame object and returns the variable
# names from the columns in the data frame.
names(text_data)

## [1] "Group"      "Baseline"   "Six_months" "Participant"
```

```
# The output confirms that the variable names include "Group", "Baseline",
# "Six_months", and "Participant".
```

```
# The command `head` prints the first 5 observations from the data frame object  
# in the console window.
```

```
head(text_data)
```

```
##   Group Baseline Six_months Participant  
## 1     1       52         32           1  
## 2     1       68         48           2  
## 3     1       85         62           3  
## 4     1       47         16           4  
## 5     1       73         63           5  
## 6     1       57         53           6
```

```
# The output confirms that all variables contain numeric values. It also  
# confirms that the data is in a wide format.
```

```
# The data is structured such that each row is an observation. The command  
# `nrow` calls the data frame object and returns the number of rows in  
# the dataset.
```

```
nrow(text_data)
```

```
## [1] 50
```

```
# The output confirms there are 50 rows in the dataset.
```

```
# The groups are coded as 1 or 2. Having the labels stored as characters is  
# helpful for making the outputs and figures readable and legible. Input  
# arguments for the function `factor` are the data frame column with group  
# coding, defined levels of 1 and 2, and labels for each level.
```

```
text_data$Group_Label <- factor(text_data$Group, levels=c(1:2),  
                                labels=c("Group 1",  
                                          "Group 2"))
```

```
# The `is.factor` command is run to confirm that Group_Label  
# is categorical.
```

```
is.factor(text_data$Group_Label)
```

```
## [1] TRUE
```

```
# TRUE is returned which confirms that the variable is categorical.
```

```
# The `is.numeric` command is to confirm that Group, Baseline, Six_months,  
# and Participant are numeric.
```

```
is.numeric(text_data$Group)
```

```
## [1] TRUE
```

```
# TRUE is returned which confirms that the variable is numeric.
```

```
is.numeric(text_data$Baseline)
```

```
## [1] TRUE
```

```
# TRUE is returned which confirms that the variable is numeric.
```

```
is.numeric(text_data$Six_months)
```

```
## [1] TRUE
```

```
# TRUE is returned which confirms that the variable is numeric.
```

```
is.numeric(text_data$Participant)
```

```
## [1] TRUE
```

```
# TRUE is returned which confirms that the variable is numeric.
```

```
# The data is reshaped to a long format using the function `melt` from the  
# package reshape2. It is stored as a new object called `long_text_data`.
```

```
long_text_data <- melt(text_data, id.vars=c("Participant", "Group_Label"),  
  measure.vars=c("Baseline", "Six_months"),  
  variable.name="Time", value.name="Value")
```

```
# For the Time variable, the string "Six_months" is changed to "Six Months" to  
# improve legibility of figures that will use this label.
```

```
long_text_data$Time <- as.factor(gsub("Six_months", "Six Months",  
  long_text_data$Time))
```

```
# The command `names` calls the data frame object and returns the variable  
# names from the columns in the data frame.
```

```
names(long_text_data)
```

```
## [1] "Participant" "Group_Label" "Time" "Value"
```

```
# The output confirms that the variable names include "Participant",  
# "Group_Label", "Time", and "Value".
```

```
# The command `head` prints the first 5 observations from the data frame object  
# in the console window.
```

```
head(long_text_data)
```

```
## Participant Group_Label Time Value  
## 1 1 Group 1 Baseline 52  
## 2 2 Group 1 Baseline 68  
## 3 3 Group 1 Baseline 85  
## 4 4 Group 1 Baseline 47  
## 5 5 Group 1 Baseline 73  
## 6 6 Group 1 Baseline 57
```

```
# The output confirms that the data is now in a long format.
```

```
# The `is.factor` command is to confirm that Group_Label and Time are  
# categorical.
```

```
is.factor(long_text_data$Group_Label)
```

```
## [1] TRUE
```

```
# TRUE is returned which confirms that the variable is categorical.
```

```
is.factor(long_text_data$Time)
```

```
## [1] TRUE
```

```
# TRUE is returned which confirms that the variable is categorical.
```

```
# The `is.numeric` command is to confirm that Participant and Value  
# are numeric.
```

```
is.numeric(long_text_data$Participant)
```

```
## [1] TRUE
```

```

# TRUE is returned which confirms that the variable is numeric.
is.numeric(long_text_data$Value)

## [1] TRUE

# TRUE is returned which confirms that the variable is numeric.

#####
##### Summary statistics
#####

# To compute summary statistics, the data object with a wide format is used.
# Here, the continuous variables, Baseline and Six_months, are stored in
# separate columns.

# The summary statistics for these continuous variables can be
# stratified by the categorical variable, Group_Label, with the function
# `by` from the pastecs package. The first input argument is the
# continuous variable, the second one is the categorical variable, and the
# third one specifies a function applied to the data. In this case, a
# new function is defined which combines the use of the functions `round` and
# `stat.desc`. Specifically, when the function `round` is called, the first
# input argument is the output from `stat.desc`, and the second input argument
# is the number of decimal places, which is specified as 4 here.

# Inputs to `stat.desc` include the following:
# - basic is set to TRUE to check how many observations are included
# - desc is set to TRUE to check the descriptive statistics values
# - norm is set to FALSE to suppress the normality test output

# The continuous variable, Baseline, is stratified by the categorical
# variable, Group_Label.
by(text_data$Baseline, text_data$Group_Label,
   function(x) round(stat.desc(x, basic=TRUE, desc=TRUE, norm=FALSE), 4))

## text_data$Group_Label: Group 1
##      nbr.val  nbr.null  nbr.na      min      max      range
##      25.0000   0.0000   0.0000   47.0000   85.0000   38.0000
##      sum      median      mean  SE.mean CI.mean.0.95      var
##     1621.0000   64.0000   64.8400    2.1359    4.4084   114.0567
##      std.dev  coef.var
##      10.6797    0.1647
## -----
## text_data$Group_Label: Group 2
##      nbr.val  nbr.null  nbr.na      min      max      range
##      25.0000   0.0000   0.0000   46.0000   89.0000   43.0000
##      sum      median      mean  SE.mean CI.mean.0.95      var
##     1640.0000   65.0000   65.6000    2.1672    4.4728   117.4167
##      std.dev  coef.var
##      10.8359    0.1652

# For the Baseline data, the output indicates that all 25 observations in each
# group were used to compute the descriptive statistics, since none were
# missing. The Group 1 has a mean of 64.84, median of 64, variance of 114.0567,
# and standard deviation of 10.6797. The range is 38, with a minimum of 47 and a

```

```
# maximum of 85. The Group 2 has a mean of 65.6, median of 65, variance of
# 117.4167, and standard deviation of 10.8359. The range is 43, with a minimum
# of 46 and a maximum of 89.
```

```
# The mean and median are fairly close and only differ by 0.84 and 0.6 for
# Group 1 and Group 2, respectively. Thus, using the mean to summarize
# the data is appropriate. Both groups have a similar mean and overall
# variability.
```

```
# The continuous variable, Six_months, is stratified by the categorical
# variable, Group_Label.
```

```
by(text_data$Six_months, text_data$Group_Label,
    function(x) round(stat.desc(x, basic=TRUE, desc=TRUE, norm=FALSE), 4))
```

```
## text_data$Group_Label: Group 1
##      nbr.val  nbr.null  nbr.na      min      max      range
##      25.0000   0.0000   0.0000   9.0000   78.0000   69.0000
##      sum      median      mean  SE.mean CI.mean.0.95      var
##      1324.0000  58.0000  52.9600   3.2662   6.7412   266.7067
##      std.dev  coef.var
##      16.3312   0.3084
## -----
## text_data$Group_Label: Group 2
##      nbr.val  nbr.null  nbr.na      min      max      range
##      25.0000   0.0000   0.0000  46.0000   79.0000   33.0000
##      sum      median      mean  SE.mean CI.mean.0.95      var
##      1546.0000  62.0000  61.8400   1.8821   3.8844   88.5567
##      std.dev  coef.var
##      9.4105   0.1522
```

```
# For the Six Months data, the output indicates that all 25 observations in each
# group were used to compute the descriptive statistics, since none were
# missing. The Group 1 has a mean of 52.96, median of 58, variance of 266.7067,
# and standard deviation of 16.3312. The range is 69, with a minimum of 9 and a
# maximum of 78. The Group 2 has a mean of 61.84, median of 62, variance of
# 88.5567, and standard deviation of 9.4105. The range is 33, with a minimum of
# 46 and a maximum of 79.
```

```
# The mean and median for Group 2 are fairly close and differ by 0.16. However,
# for Group 1, they differ by 5.04, which is a somewhat large gap and may
# warrant further testing for normality. Overall, using the mean to summarize
# the data is appropriate. Note that Group 1 has a smaller mean, larger
# range, and thus more variability.
```

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```
##### Boxplot
```

```
#####
```

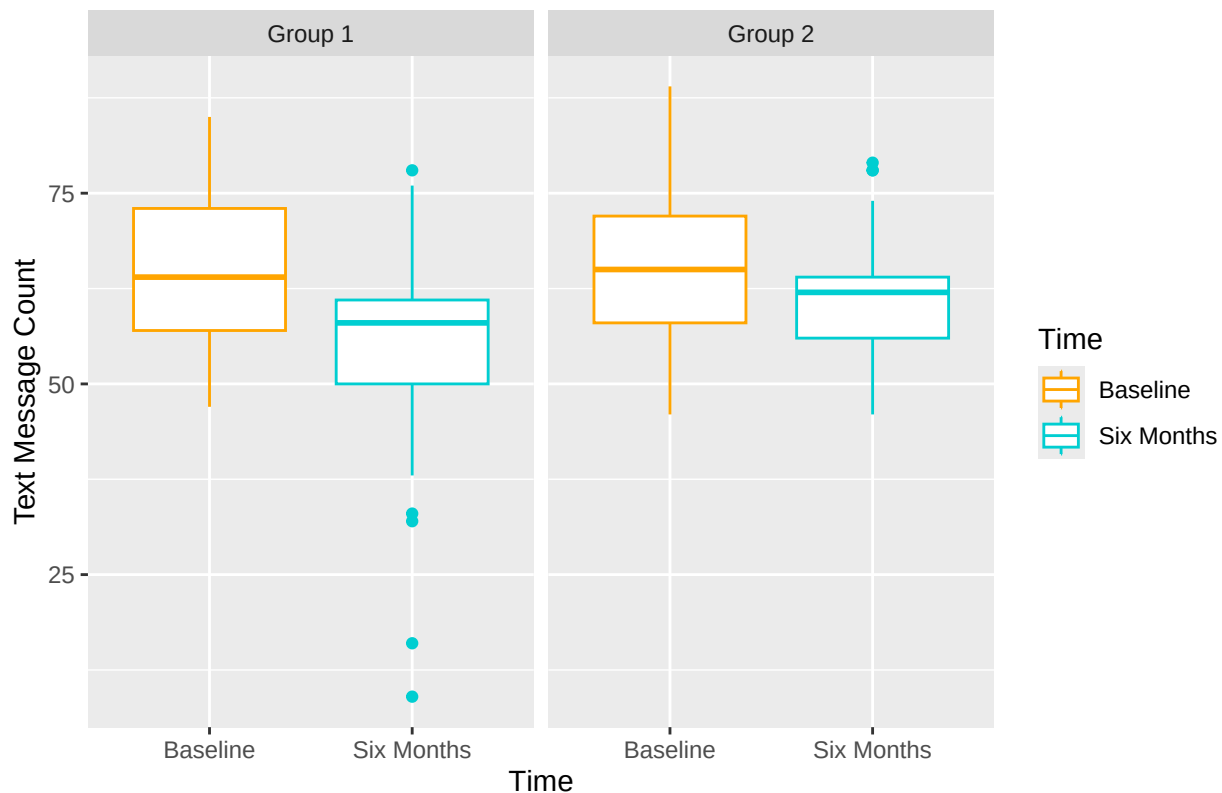
```
# To measure the difference in baseline count of text messages vs count of
# messages after 6 months, we will be employing a box plot.
# With box plots, the main "box" is comprised of the first/lower and
# third/upper quartiles, representing 25% and 75% of the data respectively.
# The line in the middle of the box is representative of the median in the data.
# Finally, the two lines extending above and below the boxes encompass all data
```

*# within 1.5x the interquartile range (IQR). Generally, the longer these are,
the greater the spread of data present. Any dots outside of these lines are
data points that are outliers.*

*# One benefit of box plots is that they can provide a visualization of the
spread of data, making it easier to spot skewness and overall distribution
shape*

```
ggplot(long_text_data, aes(x=Time, y=Value, colour=Time)) +  
  geom_boxplot() +  
  facet_wrap(~Group_Label) +  
  labs(  
    title = "Boxplot of Text Message Count by Group and Time",  
    x = "Time",  
    y = "Text Message Count") +  
  scale_color_manual(values = c("Baseline" = "orange",  
                                "Six Months" = "darkturquoise"))
```

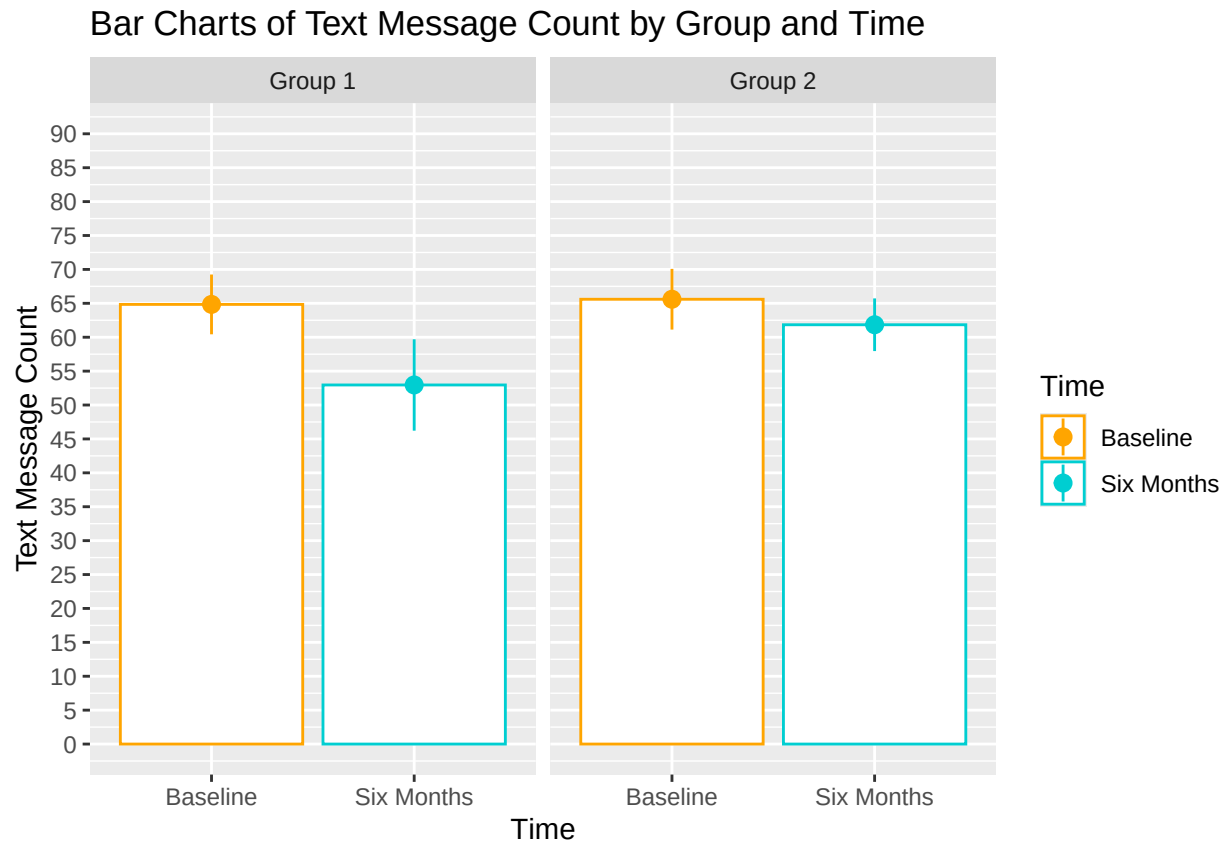
Boxplot of Text Message Count by Group and Time



*# We see that for both groups of participants, there is a decrease in the median
count of texts when comparing their baseline with their six month text count
However, it is worth noting that in both groups, the 25th percentile of the
baseline falls within the 75th percentile of the six month mark. This could
be indicative that the difference in the medians is not statistically
significant
Furthermore, we note that in the first group, there are multiple outliers,
particularly in the bottom range, which could have amounted to the
lowered median of the first group*

```
#####
##### Bar chart
#####

# A faceted bar chart for text message count by group and time is made with
# the function `ggplot` from the tidyverse package.
# Inputs to `ggplot` include the dataset object, the aesthetics function
# called `aes`, where we set the categorical variable Time to the x-axis and
# continuous variable Value to the y-axis, and assigning Time for color coding.
# This creates the first layer of the figure with the labeled axes. Next, the
# bar chart layer is created with the function `stat_summary`, which utilizes the
# mean rather than the individual observations. Inputs include defining the
# mean as the function, "bar" as the shape and white as the fill.
# Another layer is added for the error bars with the function `stat_summary`,
# which uses each group's 95% confidence interval. The function for the 95%
# confidence interval is from the package Hmisc.
# In addition, function `scale_y_continuous` is used to add another layer
# with user-defined y-axis limits. Tick marks are also specified by the input
# breaks, using y-axis limit values from 0 to 90 in increments of 5.
# Another layer is added using the function `facet_grid` to create subplots to
# separate the data by the categorical variable Group_Label, such that data
# from each group is displayed in separate columns along the x-axis. Lastly, a
# layer that specifies the axes labels and figure title is added.
long_text_data %>% ggplot(aes(x=Time,
                             y=Value, colour=Time)) +
  stat_summary(fun=mean, geom="bar", fill="White") +
  stat_summary(fun.data=mean_cl_normal, geom="pointrange") +
  scale_y_continuous(limits=c(0, 90), breaks=seq(from=0, to=90, by=5)) +
  facet_grid(. ~ Group_Label) +
  labs(title="Bar Charts of Text Message Count by Group and Time",
       x="Time",
       y="Text Message Count")+
  scale_color_manual(values = c("Baseline" = "orange",
                                "Six Months" = "darkturquoise"))
```



The bar charts show the mean value for each time and group along with error
bars, where the x-axis is time, the y-axis is the text message count, and the
subplot columns are group. In Group 1, the mean text message count decreased
from baseline to the six month time point. In Group 2, a similar pattern is
observed, however to a lesser degree. The mean and CI at baseline for both
groups are similar. However, the mean at six months appears to be smaller for
Group 1 as compared to Group 2. The CI for the six month data for Group 1 is
larger as compared to Group 2 as well.